

$$111221 \rightarrow \phi u$$

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{4}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0} [m_d^p] \delta_{i1i2} - \frac{5}{54} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1} [m_d^p] \delta_{i1i2} + \frac{16}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2} [m_d^p] \delta_{i1i2} + \right. \\
& \frac{4}{27} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0} [m_e^p] \delta_{i1i2} - \frac{5}{18} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1} [m_e^p] \delta_{i1i2} + \frac{16}{135} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2} [m_e^p] \delta_{i1i2} + \\
& \frac{2}{27} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0} [m_l^p] \delta_{i1i2} - \frac{5}{36} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1} [m_l^p] \delta_{i1i2} + \frac{8}{135} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2} [m_l^p] \delta_{i1i2} + \\
& \frac{2}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0} [m_q^p] \delta_{i1i2} - \frac{10}{58} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1} [m_q^p] \delta_{i1i2} + \frac{8}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2} [m_q^p] \delta_{i1i2} + \\
& \frac{16}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0} [m_u^p] \delta_{i1i2} - \frac{10}{27} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1} [m_u^p] \delta_{i1i2} + \frac{64}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2} [m_u^p] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} + \frac{1}{9} \mathbf{g}_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} - \frac{16}{135} \mathbf{g}_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i1}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i1}] \delta_{i1i2} - \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i1}] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i1}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i2}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i2}] \delta_{i1i2} \\
& \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i2}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i2}] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i1}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i1}] \delta_{i1i2} - \\
& \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i1}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i1}] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i2}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i2}] \delta_{i1i2} - \\
& \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i2}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i2}] \delta_{i1i2} + \\
& \frac{2}{9} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{2,2,0} [m_d^r, m_q^p] \delta_{i1i2} - \frac{1}{9} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,2,-1} [m_d^r, m_q^p] \delta_{i1i2} + \\
& \frac{2}{9} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,2,0} [m_e^r, m_l^p] \delta_{i1i2} - \frac{1}{9} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,2,-1} [m_e^r, m_l^p] \delta_{i1i2} - \\
& \frac{2}{9} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,0} [m_l^p, m_e^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,2,-1} [m_l^p, m_e^r] \delta_{i1i2} + \\
& \frac{1}{2} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{4,1,-1} [m_l^p, m_e^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{5,1,-2} [m_l^p, m_e^r] \delta_{i1i2} - \\
& \frac{2}{9} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,1,0} [m_q^p, m_d^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,2,-1} [m_q^p, m_d^r] \delta_{i1i2} + \\
& \frac{5}{6} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{4,1,-1} [m_q^p, m_d^r] \delta_{i1i2} - \frac{2}{3} \mathbf{g}_1^2 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{5,1,-2} [m_q^p, m_d^r] \delta_{i1i2} - \\
& \frac{1}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{2,2,0} [m_q^p, m_u^r] \delta_{i1i2} + \frac{1}{18} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,2,-1} [m_q^p, m_u^r] \delta_{i1i2} - \\
& \frac{1}{18} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,0} [m_q^p, \tilde{\mu}] + \frac{1}{36} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,2,-1} [m_q^p, \tilde{\mu}] + \\
& \frac{1}{36} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{3,1,-1} [m_q^p, \tilde{\mu}] + \frac{1}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,1,0} [m_u^r, m_q^p] \delta_{i1i2} + \\
& \frac{1}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,2,-1} [m_u^r, m_q^p] \delta_{i1i2} + \frac{1}{3} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{4,1,-1} [m_u^r, m_q^p] \delta_{i1i2} - \\
& \frac{2}{3} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{5,1,-2} [m_u^r, m_q^p] \delta_{i1i2} - \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0} [m_u^{i1}, m_1] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1} [m_u^{i1}, m_1] \delta_{i1i2} - \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0} [m_u^{i1}, m_3] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1} [m_u^{i1}, m_3] \delta_{i1i2} - \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0} [m_u^{i2}, m_1] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1} [m_u^{i2}, m_1] \delta_{i1i2} - \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0} [m_u^{i2}, m_3] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1} [m_u^{i2}, m_3] \delta_{i1i2} - \frac{5}{18} \mathbf{g}_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} + \\
& \frac{2}{9} \mathbf{g}_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{5}{6} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} + \frac{2}{3} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \delta_{i1i2} + \\
& \frac{1}{36} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,0} [\tilde{\mu}, m_q^p] - \frac{11}{36} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_q^p] + \\
& \frac{1}{9} \mathbf{g}_1^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_q^p] - \frac{1}{18} \mathbf{g}_1^4 \text{LF}_{2,1,1,-1} [m_1, m_u^{i1}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{9} \mathbf{g}_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_u^{i1}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{18} \mathbf{g}_1^4 \text{LF}_{2,1,1,-1} [m_1, m_u^{i2}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{9} \mathbf{g}_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_u^{i2}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{8} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^p] - \\
& \frac{1}{18} \mathbf{g}_1^4 \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i1}] \delta_{i1i2} + \frac{1}{18} \mathbf{g}_1^4 m_1^2 \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i1}] \delta_{i1i2} - \\
& \frac{1}{18} \mathbf{g}_1^4 \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i2}] \delta_{i1i2} + \frac{1}{18} \mathbf{g}_1^4 m_1^2 \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] \delta_{i1i2} + \\
& \frac{3}{8} \mathbf{g}_2^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^p] - \frac{1}{4} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_q^p] - \\
& \frac{3}{4} \mathbf{g}_2^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_2, m_q^p] - \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{1,1,1,1,0} [m_1, m_q^p, m_u^{i1}, \tilde{\mu}] + \\
& \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_1, m_q^p, m_u^{i1}, \tilde{\mu}] - \\
& \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{1,1,1,1,0} [m_1, m_q^p, m_u^{i2}, \tilde{\mu}] + \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \\
& \text{LF}_{2,1,1,1,-1} [m_1, m_q^p, m_u^{i2}, \tilde{\mu}] - \frac{1}{4} \overline{y_u^{si1}} y_u^{pi2} \overline{a_d^{pr}} a_d^{sr} \text{LF}_{2,1,1,1,-1} [m_d^r, m_q^p, m_q^s, \tilde{\mu}] - \\
& \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_u^{i1}, m_u^{i2}] - \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \\
& \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_u^{i2}, \tilde{\mu}] - \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_u^{i2}, \tilde{\mu}] - \\
& \frac{2}{3} \mathbf{g}_3^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_3, m_u^{i1}, m_u^{i2}] + \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{ps}} \overline{y_u^{ri1}} y_u^{pi2} y_u^{rs} \\
& \text{LF}_{2,1,1,1,-1} [m_u^s, m_q^p, m_q^r, \tilde{\mu}] + \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_u^{i1}, m_1, m_q^p, \tilde{\mu}] + \\
& \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [m_u^{i2}, m_1, m_q^p, \tilde{\mu}] - \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \\
& \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_q^p, m_u^{i1}] - \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{pi1}} y_u^{pi2} \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_q^p, m_u^{i2}] \Big)
\end{aligned}$$